



PRATT NSIAH BOSSMAN

Robotics Engineering Student • AI-Driven Developer

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PROFILE

Third-year Senior High School student studying **Robotics Engineering** at **AWASO STEM High School**. I am an AI-driven developer who specializes in leveraging machine learning to improve accuracy, efficiency, debugging, analysis, and research across my work. I am proficient in **HTML, CSS, Python, and C++**, with hands-on experience in **Tkinter, Arduino, SDL, and OpenGL** for 3D graphics. I am currently building **CloudTok** (a full-stack social platform inspired by TikTok) and **MotionForge** (a 3D motion-capture and animation tool), with the long-term goal of creating innovative, AI-powered gaming experiences.

EDUCATION

AWASO STEM High School	2024 – Present
Robotics Engineering — Senior High School (3rd Year)	

TECHNICAL SKILLS

Languages	HTML, CSS, JavaScript, Python, C++
Frameworks & Libraries	React, Vite, TypeScript, Tailwind CSS, shadcn/ui, Three.js, Tkinter, SDL2, OpenGL (GLAD/GLM), Express, Flask
AI & ML	MediaPipe (pose/motion), Stability AI, Pollinations, OpenRouter LLM APIs, computer vision, ML-assisted debugging & analysis
Hardware & Embedded	Arduino (Firmata, Servo, TFT, WiFinINA), HC-SR04 ultrasonic sensor, serial (pyserial) communication
Tools & Platforms	Git, Cloudflare Workers / D1 / R2, WebRTC, PWA / Service Workers, IndexedDB, JSON, SymPy, NumPy, Matplotlib

PROJECTS

Flagship & Currently Developing

CloudTok — Full-Stack Short-Video Social Platform

HTML, CSS, JavaScript • Cloudflare Workers • D1 (SQL) • R2 • WebRTC

A TikTok-inspired social platform built to master modern web and cloud engineering. Features video upload, feed and player, authentication with an auth guard, likes/comments/saves, a follow system, hashtag search, real-time chat with WebRTC calling and live streaming, a payments/gifts system, an admin panel, and AI thumbnail generation and recommendations.

MotionForge — Web 3D Motion-Capture & Animation Tool

Vite • Three.js (r160) • MediaPipe Pose • PWA / IndexedDB • JavaScript

A mobile-first, installable PWA that turns a webcam into a motion-capture studio. Imports GLB/GLTF/FBX/OBJ models, performs skeleton auto-rigging and rig-agnostic motion retargeting via MediaPipe Pose, and exports baked animations to GLB. Includes offline support and touch controls.

AI Tools

3D-Model-AI — Text-to-3D Model Generator

Node.js • Express • Three.js • Stability AI (image → GLB mesh)

Full-stack tool that converts a text prompt into a 3D model: it generates a 2D image then a 3D GLB mesh, with an in-browser orbit-controlled viewer and export to GLB/GLTF/OBJ/STL.

Video Gen AI (Script2Video Studio)

HTML/CSS/JS • Python (yt-dlp, imageio-ffmpeg, LLM APIs) • OpenRouter

Hybrid tool that turns a written script into a finished video — each paragraph becomes a timed scene with auto-sourced footage, voiceover (TTS) and optional background music.

AI Image Generator

HTML • CSS • JavaScript • Pollinations.ai API

Web app that generates images from prompts with style presets (Cinematic, 3D Render, Anime, Cartoon, Fantasy, Product), produces two variations, supports regeneration, download, and localStorage history.

3D Video & 2D Animation AI Studio

React 18 • Vite • TypeScript • Tailwind • shadcn/ui

A modern, component-rich UI studio prototype for generating 3D videos and 2D animations from text prompts, with Generate/History/Gallery/Settings workflows.

Almost Real AI — Desktop Assistant

Python • Tkinter • SymPy • NumPy • Matplotlib • PIL

A Tkinter AI assistant with persistent JSON memory, written-number-to-math conversion, safe expression evaluation, algebraic equation solving, and result plotting.

Games & Game Engines

3D Fighting Game — Custom C++ Engine

C++ • SDL2 • OpenGL/GLAD • GLM • assimp • Dear ImGui

A from-scratch 3D fighting game engine with arena and open-world (GTA-style) modes, a hitbox/combat system, HUD, an in-editor mouse gizmo, audio/diagnostics subsystems, and bundled 3D assets and shaders.

Zeal of Conquest

HTML/CSS/JS (web) • SDL2 (C++ prototype)

A strategy-game menu and UI with a campaign map (Rome, Greece, Alps, Egypt), settings (volume/graphics), and credits, plus an early SDL2 renderer prototype.

SDL Game & Pratt Engine

C++ • SDL2 • OpenGL

A delta-time SDL2 + OpenGL animated-graphics demo (sine-wave triangle, input handling) and a minimal SDL2 window/event-loop engine skeleton.

Web Apps & Websites

PRANSBO TECH — Personal Portfolio

HTML • CSS • JavaScript

A responsive portfolio with project showcase, dark/light theme toggle, scroll-progress bar, and Open Graph / Twitter cards.

ReggieK Enterprise

HTML • CSS • JavaScript • JSON-LD

A natural health-supplements brand site with structured data (schema.org), Open Graph tags, and GitHub Pages deployment.

Aunty's Logistics Website (SwiftLine Logistics)

HTML • CSS • JavaScript

A business site with a services grid (Air/Sea/Road freight, warehousing), shipment-tracking form, and a validated quote/contact form.

Mediamega

HTML • CSS • JavaScript • Paystack

An entertainment/media portal (movies, songs, games, software, comedy) with client-side search, profile modals, Paystack payments, and an enterprise product sub-site.

WhatsApp Interface Webapp

HTML • CSS • JavaScript

A WhatsApp-style chat UI clone with contact sidebar, message bubbles, and switchable conversations.

Scott's Web & Template Portals (1 & 2)

HTML • CSS • JavaScript • JSON (PHP experiments)

A personal poetry site with audio playback, plus two entertainment-portal templates (movies/songs/games) with search and sign-in/sign-up flows and early PHP backend experiments.

Hardware & Embedded

Simp Arduino & Arduino LED Controller

Arduino (C/C++) • Firmata/Servo/TFT/WiFiNINA • Python + Tkinter + pyserial

An Arduino workspace bundling standard hardware libraries (servo, TFT, WiFi, ultrasonic sensors) and a Python Tkinter GUI that toggles an LED on an Arduino over COM3.

Python Desktop Utilities

Simpltkint Framework & Practice

Python • Tkinter

A lightweight Tkinter GUI framework (wrapper helpers for windows, buttons, labels, entries, dialogs) with a usage example, plus C++ OOP battle-simulation practice (classes, pointers, dynamic arrays).

REFERENCES

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